

Characterization of Multi-Path Interference of 10-km Hollow Core Fiber using Swept Wavelength Interferometry

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Abstract: We measure the impulse response of a 10-km support-tube hollow core fiber (ST-HCF) showing the effects of higher-order mode propagation and multi-path interference. A linear split-step simulation validates both the measurement and model.

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1. Introduction

Multi-path interference (MPI) is the coherent beating between weak time-delayed replicas traveling on separate spatial paths that coherently beat with the desired signal. Fiber physics will produce MPI and some examples include higher-order mode (HOM) propagation in the core and cladding, polarization mode dispersion, and double Rayleigh scattering in amplified Raman systems [1]. Humans introduce MPI from installation errors such as tight bends, multi reflections between splice points and dirty connectors [2], HOM propagation in bend insensitive fibers [3], and devices relying on higher order modes. In datacenters and complex optical networks [3], where there are many numerous connectors and unusual fiber routing, MPI is a dominant and hard to troubleshoot impairment [4]. In quasi-single mode fiber transmission in multimode fibers [5], MPI results from mode coupling between the fundamental mode and HOMs. Effectively, HCF is a quasi-single mode fiber (e.g., HOMs co-propagate, but have high loss) and suffers from MPI.

The MPI problem is worse for ultra-low loss HCF, such as the ST-HCF used here [6], where the goal would be 0.05 dB/km loss for the fundamental mode and 100s dB/km for HOMs. Low-loss HCF requires relatively large cores - approximately 4 times bigger than SMF. The dominant loss source in low-loss HCF is leakage loss from mode coupling to HOMs and eventually out of the fiber [7]. The mode coupling process is always reciprocal, that is, the power coupled between the fundamental and a HOM is the same as between the HOM and the fundamental. The coupling back and forth produces the MPI. Clever fiber design to minimize MPI attempts to maximize the HOM loss while minimizing the fundamental mode loss. One can slightly increase the leakage loss by using a smaller core which will have a much larger increase in HOM loss [7, 8]. However, the low loss HCF (e.g., 0.1 dB/km vs. 0.2 dB/km) tends to have lower loss for the HOMs too (e.g., 3 dB/km compared to 30 dB/km). Two additional sources of MPI in HCF are from mode conversion at splice points, and from strong fresnel reflections between HCF and SMF (a 4% reflection between air and glass).

The two main methods to measure MPI [9] are the: scanning wavelength method where the power variance across the spectrum is measured with sufficient spectral resolution, and the output power variance technique where a narrowband laser is set to a particular wavelength and the power fluctuations are measured over time. These

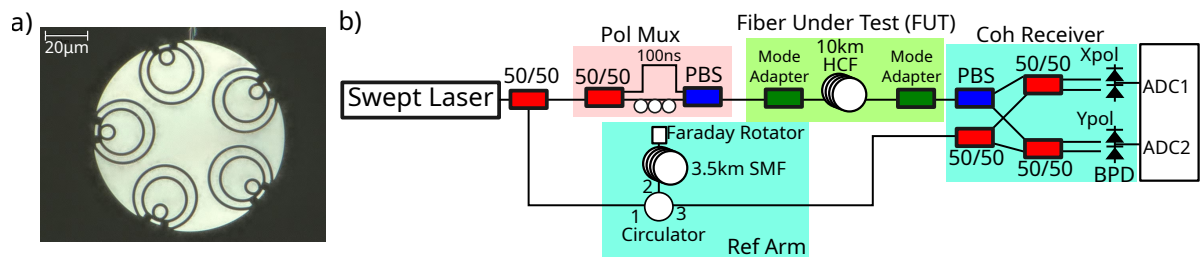


Fig. 1: a) Cross section of ultra-low loss support tube hollow core fiber (ST-HCF). b) Diagram of swept wavelength interferometer.

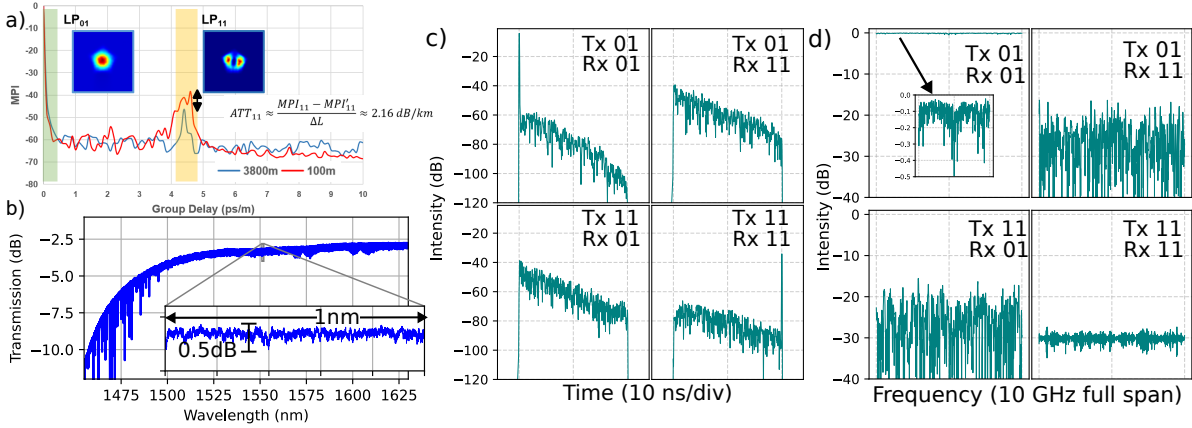


Fig. 2: a) S^2 measurement of the HCF. b) Pure intensity transmission of the fiber under test (HCF+mode adapters+connectors). c,d) Simulation of transfer matrix of a 10-km two-mode fiber with DGD of 5 ns/km, and 0 dB/km LP_{01} mode loss and 3 dB/km LP_{11} mode loss). Ideally, the Tx01-Rx01 cell is only excited by the mode adapters.

methods infer the MPI from intensity fluctuations. Spatial and spectral imaging (S^2) techniques [10] can capture modal delays and some limited information about MPI. Here, we measure the temporal impulse response produced by the MPI as seen by the fundamental mode using a coherent swept-wavelength interferometer (SWI) [11–14], which captures the amplitude and phase transfer function of the HCF precisely over 150 nm.

2. Experiments and Results

Figure 2a) shows a S^2 measurement of the HCF for two different lengths of fiber and provides reference values for the differential group delay (DGD) of 4.5 ns/km and a differential mode attenuation (DMA) of 2.16 dB/km. Figure 2b) shows an intensity transmission measurement of the fiber under test (FUT) with mode adapters as indicated in Fig. 1b). A separate OFDR [13] measured about 0.12 dB/km loss at 1550 nm and the loss of the 10-km FUT of 3 dB is consistent with additional losses from the mode adapters [15]. Note, from the spectral intensity fluctuations the MPI is around -50 dB/km (formula 4 in [9]).

A linear split-step simulation shown in Fig. 2(c,d) of a two mode fiber depicts the expected impulse responses and spectral transmission. At each 0.5 m step energy couples between the two modes for -50 dB/km MPI. S^2 and intensity measurements do not capture the phase response and therefore cannot capture the impulse response of the fiber which would be extremely useful for modeling using DSP. Note, the FUT transfer matrix cell is the Tx01-Rx01 cell which means launch on the fundamental mode and receive on the fundamental mode. This assumes perfect mode adapters, and quite often a HOM will be launched at the input, and a significant amount of energy could propagate via the HOM and couple into the output SMF via the output mode adapter. Splice points can couple energy between the LP_{01} and LP_{11} modes, which means that MPI arising from these HOMs is a critical consideration, especially when using shorter fiber spans and jumpers.

Figure 1b) shows the SWI [12, 14] used to measure the HCF's transfer function where we expect a result similar to the simulations. A mode-hop free laser (EXFO T200S) sweeps between 1460 nm and 1630 nm at 200-nm/s and a low-speed coherent detector (100 MS/s) records the interference between the SMF reference arm (PMD compensated with Faraday rotator) and the FUT. 0-dBm is launched into the FUT and the reference arm. The laser's lorentzian linewidth is below 10-KHz which helps to see weak signals in the impulse response. The FUT delay is always greater than the reference arm delay which ensures that the amplitude and phase spectral transmission can be unambiguously extracted from the spectral interference using only 1 BPD per polarization (e.g., Heterodyne detection). A Fourier transform of the spectral transmission produces temporal impulse responses. It also resolves the four Jones elements by launching two orthogonal delayed polarization replicas and receiving both polarizations. Full details are in [12, 13].

Figure 3a) shows the averaged intensity impulse response from 1470 nm to 1620 nm over the four Jones elements. We can identify three features: the fundamental mode (red), the MPI plateau (green), and the light traveling completely on the HOM due to imperfect SMF to HCF mode adapters (blue). The inset identifies that very close to the fundamental mode peak there is an additional steep slope. This is most likely MPI from a higher order radially symmetric LP_{0n} mode that has significantly larger loss than the LP_{11} but has stronger coupling to the fundamental mode due to symmetry considerations. Integrating the total energy in the MPI plateau (green) and comparing to the power in the fundamental mode (red) indicates an MPI of around 53 dB/km.

Figure 3b) shows this impulse response spectrally resolved across the entire wavelength range. At longer wavelengths, the HOM energy is much stronger than at shorter wavelengths as indicated by the HOM mode peak. The HCF loss is higher at shorter wavelengths and therefore the HOM loss would be proportionally much larger. Fig-

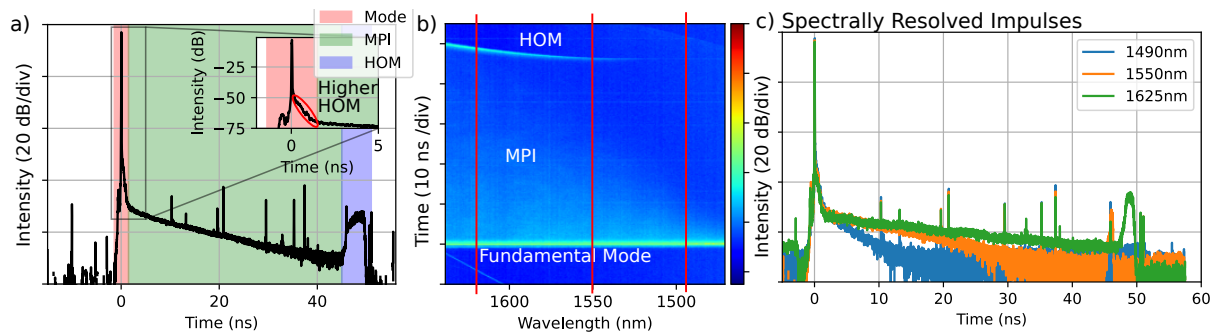


Fig. 3: Swept Wavelength Interferometer measurements averaged over 4 Jones elements. a) Time domain impulse response. b) Time domain impulse. c) Impulse response at three wavelengths.

ure 3c) shows impulse responses at three different wavelengths. 1625 nm contains the largest HOM component and also shows the most power in the MPI plateau (MPI 50 dB/km) and 1490 nm shows the least MPI as indicated by the MPI plateau falling directly into the noise floor due to the higher loss at 1490 nm (MPI 55 dB/km).

In conclusion, we used SWI to measure the impulse response of a HCF over 150 nm of bandwidth. This is the first measurement of the impulse response of a low-loss HCF, showing the MPI across wavelength, and matching well with a simple split-step model. These results can help DSP engineers create novel algorithms for future ultralong haul HCF networks.

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